

**Eighth Semester B. E. (Electronics and Communication Engineering)
Examination****REAL TIME OPERATING SYSTEMS AND KERNELS**

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
- (2) All questions carry marks as indicated.
- (3) Illustrate your answer with neat sketches.
- (4) Assume suitable data wherever necessary.

1.
 - (a) Describe the differences between symmetric and asymmetric multiprocessing. What are three advantages and one disadvantage of multiprocessor systems ? 5(CO1)
 - (b) What are the two models of inter-process communication ? What are the strengths and weaknesses of the two approaches ? 5(CO1)
2.
 - (a) What is dispatcher, explain with functions and features ? 5(CO2)
 - (b) Consider the following set of processes, with the length of the CPU burst given in milliseconds :

Process	Burst Time	Priority
P1	2	2
P2	1	1
P3	8	4
P4	4	2
P5	5	3

The processes are assumed to have arrived in the order P1, P2, P3, P4, P5 all at time 0 :

- (a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms : FCFS, SJF, non-preemptive priority (a larger priority number implies a higher priority) and RR (quantum = 2).

- (b) What is the turnaround time of each process for each of the scheduling algorithms in part (a) ?
- (c) What is the waiting time of each process for each of these scheduling algorithms ?
- (d) Which of the algorithms results in the minimum average waiting time (over all processes) ? 5(CO2)
3. (a) Explain with an example a situation in which pipes are more suitable than named pipes and an example of a situation in which named pipes are more suitable than ordinary pipes. 5(CO3)
- (b) What are the advantages and disadvantages of supporting memory mapped I/O to device control registers ? 5(CO3)
4. (a) Consider the following segment table :

Segment	Base	Length
0	219	600
1	2300	14
2	90	100
3	1327	580
4	1952	96

What are the physical addresses for the following logical addresses ?

- (a) 0,430
- (b) 1,10
- (c) 2,500
- (d) 3,400
- (e) 4,112

5(CO4)

- (b) A certain computer provides its users with a virtual memory space of 232 bytes. The computer has 222 bytes of physical memory. The computer has 222 bytes of physical memory. The virtual memory is implemented by paging, and the page size is 4,096 bytes. A user process generates the virtual address 11123456. Explain how the system establishes the corresponding physical location. 5(CO4)
5. (a) Consider a system that supports the strategies of contiguous, linked and indexed allocation. What criteria should be used in deciding which strategy is best utilized for a particular file ? 5(CO4)
- (b) None of the disk-scheduling disciplines, except FCFS, is truly fair (starvation may occur). Explain why this assertion is true. 5(CO4)
6. Summarize computation of performance benchmarks of Real Time Systems. 10(CO5)

